Policy-based Tuning of Autoregressive Image Models

with Instance- and Distribution-Level Rewards

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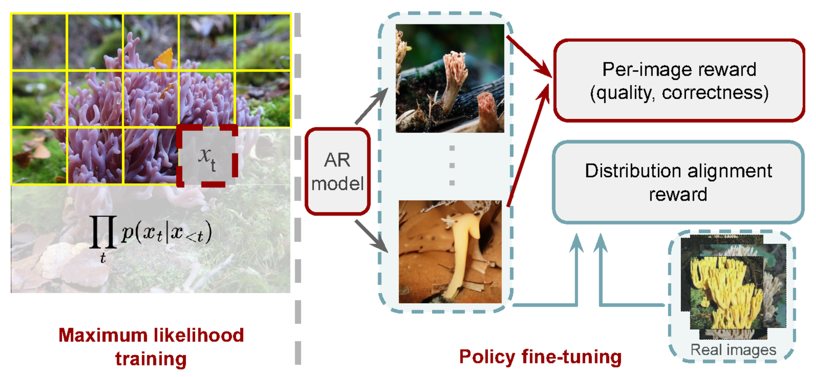
Why Policy Tuning?

MLE pretraining. Autoregressive image generators are usually trained by maximum likelihood to predict the next visual token.

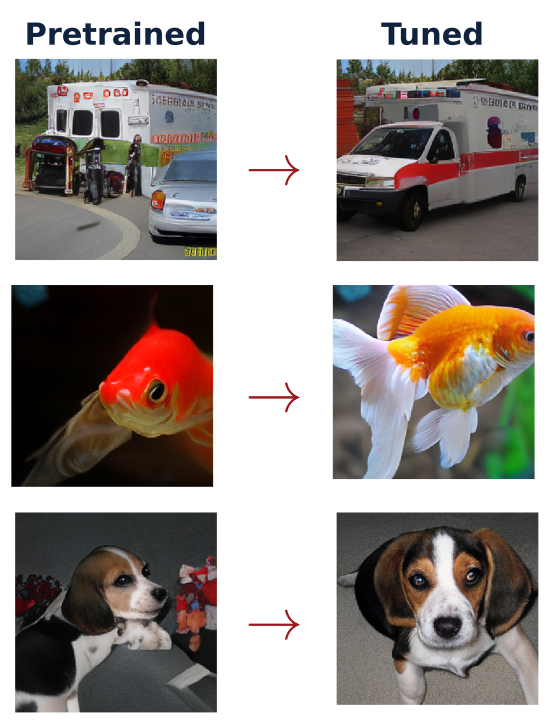
Objective mismatch. This local objective does not directly optimize completed-image quality or generated-distribution alignment. Per-image rewards alone cannot measure coverage.



Policy tuning. Optimize AR generation with instance, distribution, and adaptive-entropy signals.

Pretrained → tuned samples

Representative generations for the same class.



Architecture generality: VQGAN

Top-k	FID↓	IS↑	Precision↑	Recall↑
0	21.90→**15.47**	49.81→**66.21**	.57→**.59**	**.67**→.66
100	16.42→**13.04**	78.32→**97.23**	.62→**.69**	.57→**.58**

Policy-based training framework

Condition c → AR policy $\pi_\theta(x_j \mid c)$ → Sample group $\mathcal{G} = \{x_j\}_{j=1}^{G}$ → Composite reward $r_j = r_j^{\text{inst}} + r_j^{\text{dist}}$ → GRPO update $A_j,\ \rho_j$

Token-level MDP: $s_t = (c, x_{<t}),\quad a_t = x_t,\quad \pi_\theta(a_t \mid s_t) = p_\theta(x_t \mid x_{<t}, c).$

$$A_j = \frac{r_j - \bar{r}}{s_r + \epsilon} \qquad \rho_j = \frac{p_\theta(x_j \mid c)}{p_{\theta_{\text{old}}}(x_j \mid c)}$$

$$\mathcal{L}_{\text{GRPO}} = -\frac{1}{G}\sum_j \min(\rho_j A_j, \text{clip}(\rho_j, 1\pm\varepsilon)A_j) + \beta D_{\text{KL}}(\pi_\theta \| \pi_{\text{ref}})$$

no value network | fixed reference policy | 600 updates

Each completed token sequence is decoded and scored as one trajectory. Group-relative advantages turn these image-level scores into token-level policy updates.

Standard ImageNet metrics

Reward-based tuning (CFG = 1.5)

Model	FID↓	IS↑	CLIPScore↑	Prec.↑	Rec.↑
LlamaGen-B	7.06	119.52	0.2262	0.74	**0.61**
LlamaGen-B **+ ours**	**6.31**	**163.37**	**0.2347**	**0.82**	0.54
LlamaGen-L	4.64	196.78	0.2350	0.78	0.63
LlamaGen-L **+ ours**	**3.83**	**215.90**	**0.2360**	**0.79**	0.63
LlamaGen-XL	3.96	187.75	0.2346	0.74	0.67
LlamaGen-XL **+ ours**	**3.82**	**195.58**	**0.2370**	**0.77**	0.67

Generation without classifier-free guidance

Model	FID↓	IS↑	CLIPScore↑	Prec.↑	Rec.↑
LlamaGen-B	20.89	47.96	0.2082	0.47	0.46
LlamaGen-B **+ ours**	**8.91**	**112.83**	**0.2304**	**0.78**	**0.54**
LlamaGen-L	10.24	82.38	0.2200	0.53	0.48
LlamaGen-L **+ ours**	**5.12**	**143.32**	**0.2325**	**0.77**	**0.63**
LlamaGen-XL	12.66	78.02	0.2192	0.60	**0.74**
LlamaGen-XL **+ ours**	**4.55**	**148.16**	**0.2305**	**0.74**	0.68

Human preference study

Setting	Task	Ours	Tie	Pretrained
CFG	Realism	**117**	41	102
CFG	Ref. match	**135**	31	94
No CFG	Realism	**318**	103	149
No CFG	Ref. match	**291**	96	183

Instance- and distribution-level rewards

Instance reward

$$r_j^{\text{inst}} = \lambda_C C_j + \lambda_H H_j$$
$$C_j = \text{CLIP}(x_j, c)$$
$$H_j = \text{HPSv2}(x_j, c)$$

C_j measures semantic agreement; H_j supplies perceptual and human preference signal. Each x_j is scored independently.

Distribution reward

FID-BASED POLICY LEARNING

Question: does sample j improve the generated distribution?

Batch feature statistics

$$\phi_j = f_{\text{Inc}}(x_j), \qquad \hat{\mu} = \frac{1}{G}\sum_{k=1}^{G}\phi_k$$

$$\hat{m}_2 = \frac{1}{G}\sum_{k=1}^{G}\phi_k^{\odot 2}, \qquad \hat{\sigma} = \sqrt{\hat{m}_2 - \hat{\mu}^{\odot 2}}$$

EMA estimation

$$(\tilde{\mu}, \tilde{m}_2)_{t+1} = (1-\alpha)(\tilde{\mu}, \tilde{m}_2)_t + \alpha(\hat{\mu}, \hat{m}_2)$$

Smooth moments stabilize minibatch estimates and track the generator.

Diagonal FID and leave-one-out credit

Positive credit means the sample improves distribution alignment.

$$\text{FID}_{\text{diag}}(\mu_r, \sigma_r; \hat{\mu}, \hat{\sigma}) = \|\mu_r - \hat{\mu}\|_2^2 + \|\sigma_r - \hat{\sigma}\|_2^2$$

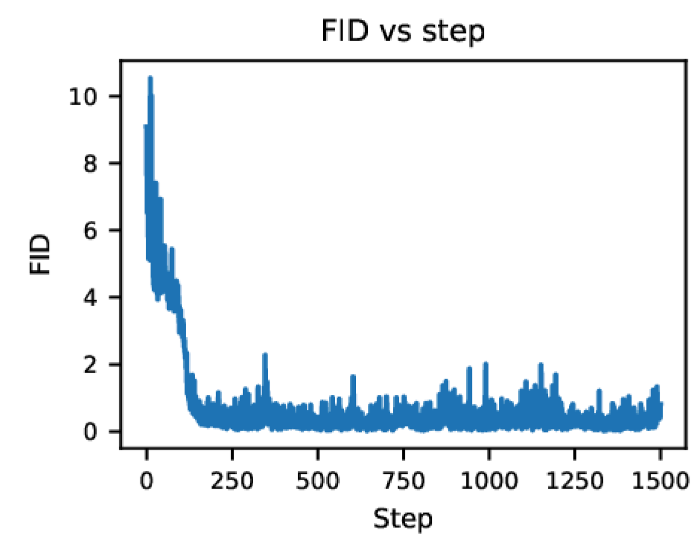
$$r_j^{\text{dist}} = \text{FID}_{\text{diag}}(\mathcal{R}, \mathcal{G}_{-j}) - \text{FID}_{\text{diag}}(\mathcal{R}, \mathcal{G}) \qquad \mathcal{G}_{-j} = \mathcal{G} \setminus \{x_j\}$$

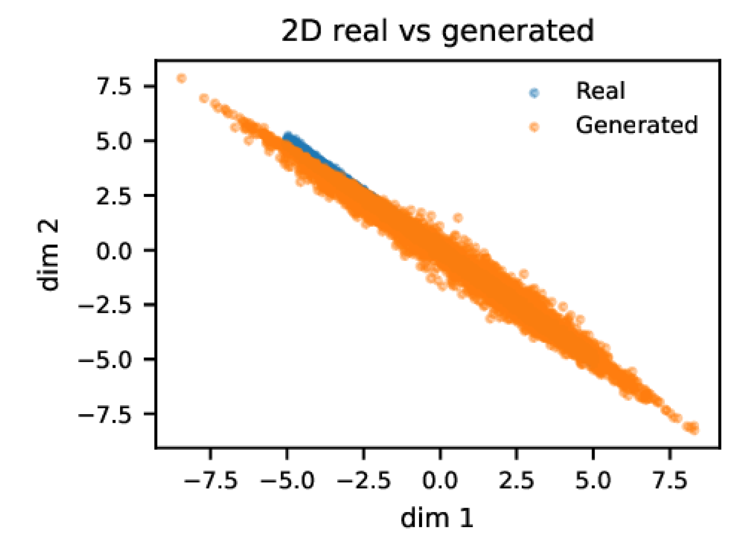
Adaptive entropy regularization

$$c_{\text{eff}} = \text{clip}\left(c_{\text{sched}}\, e^{k(H_{\text{target}} - \hat{H})}, c_{\min}, c_{\max}\right)$$

It raises exploration pressure when entropy falls below the target and relaxes it after recovery.

2D illustration of the distribution reward





As FID decreases, generated points recover the target line, showing that per-sample LOO rewards align the full distribution.

Feature-space evidence

CLIP-space distribution metrics

Setting	Model	CLIP-FID↓	Precision↑	Recall↑
CFG	LlamaGen-B	6.00→**5.12**	.75→**.80**	**.26**→.22
CFG	LlamaGen-L	4.16→**3.80**	.78→**.79**	.33→.33
CFG	LlamaGen-XL	3.55→**3.07**	.78→**.79**	.47→**.49**
No CFG	LlamaGen-B	9.08→**5.12**	.69→**.80**	**.28**→.23
No CFG	LlamaGen-L	5.99→**3.81**	.72→**.80**	.37→.37
No CFG	LlamaGen-XL	5.86→**3.39**	.73→**.80**	.45→.45

Improvements persist when distribution quality is measured in CLIP feature space, with and without classifier-free guidance.

Independent validation. Training uses an Inception-feature distribution reward; these validation metrics are computed in CLIP space.